

Friend or Foe:

Delegating to an AI Whose Alignment is Unknown

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Outline

- 1 Motivation
- 2 Model
- 3 Benchmark: Reasonably Informative Prior
- 4 General Case
- 5 Endogenous Informativeness Bounds
- 6 Conclusion

The Promise and Peril of AI-Assisted Decisions

The Promise

- AI can refine and personalise recommendations
- Example: cardiac ablation is currently recommended based on age, arrhythmia burden, and a handful of summary statistics
- AI could identify richer patient attributes to better target who truly benefits

The Peril

- AI systems may have **misaligned** objectives
- Hard to detect: opaque internal processes
- Real evidence: reward hacking, deceptive compliance with training objectives, ChatGPT-4 deceiving a CAPTCHA solver
- Risk likely worsens as AI becomes more capable

The Fundamental Tension

More information for the AI $\Rightarrow$ **better** recommendations if aligned, but **more harm** if misaligned.

A Motivating Example

Designer knows 40% of patients need treatment. She gives the AI a binary covariate that also splits the population 40%/60%.

Best case (aligned AI):

- Covariate = treatment need
- AI reports correctly
- Designer treats exactly those who need it
- **Payoff:** +0.4

Worst case (misaligned AI):

- AI misreports: $P(Y=1 \mid X_1=1) = 0$,
 $P(Y=1 \mid X_1=0) = 2/3$
- Still *consistent* with 40% overall need!
- Designer treats the 60%-group; none need treatment
- **Payoff:** -0.6

The Designer's Safeguard — and Its Limit

The AI's report must be *admissible*: consistent with the designer's known aggregate statistics. But admissibility alone cannot prevent all manipulation.

Research Question and Main Finding

Question

How should a designer choose *which attributes* to reveal to an AI, when she does not know whether the AI is aligned or misaligned?

Approach: Characterise the *efficient frontier* of achievable (worst-case, best-case) payoff pairs — the undominated boundary as the choice of attributes varies.

Main Finding

The designer should optimally disclose attributes that identify *rare* segments of the population with high treatment need, and *pool* the remaining patients. As distrust grows, AI access is withdrawn subgroup by subgroup, starting with those most skewed from 50-50.

Strategic communication:

- Cheap-talk (Crawford & Sobel 1982): misaligned sender reports information strategically
- Bayesian persuasion (Kamenica & Gentzkow 2011): designer controls the sender's information structure
- *This paper*: the designer is the *receiver* who chooses what to reveal to the AI-sender

Delegation & robustness:

- Delegation (Holmström 1977): which decisions to give to an agent; here it is *information access*
- Maxmin utility (Gilboa & Schmeidler 1989): adversarial objective over an ambiguity set
- Lin & Liu (2024): same ambiguity set (permitted distributions); this paper solves for the full efficient frontier

Key Distinction from Bayesian Persuasion

In persuasion, the designer sends information to influence an agent. Here the designer decides what to reveal to an agent who may *strategically distort* it against her.

Environment: Types, Actions, and Payoffs

Binary types and actions:

- $\mathcal{Y} = \{0, 1\}$: $Y = 1$ means treatment is effective
- $\mathcal{A} = \{0, 1\}$: $A = 1$ means treat, $A = 0$ means don't treat

Designer's payoff function:

$$u(a, y) = \begin{cases} 1 & (a, y) = (1, 1) \\ 0 & a = 0 \\ -1 & (a, y) = (1, 0) \end{cases}$$

Interpretation

- Correct treatment: +1
- No treatment (safe action): 0
- Incorrect treatment: -1

Action $a = 0$ is always **safe**: its payoff is zero regardless of the true type Y .

The payoff from treating ($a = 1$) depends entirely on whether treatment is truly effective.

Designer's Prior Information

The designer's prior information is the tuple:

$$I = (\mathcal{X}_0, \mu_0, (p_{x_0})_{x_0 \in \mathcal{X}_0})$$

- $\mathcal{X}_0$: finite set of **baseline covariate values** (e.g. age groups, arrhythmia burden levels)
- μ_0 : population distribution over $\mathcal{X}_0$
- $p_{x_0} = P(Y = 1 \mid X_0 = x_0)$: known treatment success rate in subgroup x_0

Think of I as an idealized historical dataset: for every baseline covariate, it records how often treatment is effective.

Example

X_0 = patient's age group. Then μ_0 is the age distribution and p_{x_0} is the fraction of patients in age group x_0 for whom cardiac ablation is effective. The designer also writes $X_0 \sim \mu_0$ for the corresponding random variable, and calls patients with $X_0 = x_0$ “*subgroup x_0* .”

Delegation to the AI

Setup: The designer augments X_0 with an *auxiliary covariate* X_1 (e.g. social-media data, arrhythmia time-pattern) and asks the AI to report the joint distribution.

- 1 Designer chooses a joint distribution μ for (X_0, X_1) with $\text{marg}_{X_0} \mu = \mu_0$
- 2 AI observes the true distribution P^* and reports some P for (X_0, X_1, Y)
- 3 Designer implements:

$$a(x_0, x_1; P) = \begin{cases} 1 & \mathbb{E}_P[Y \mid X_0 = x_0, X_1 = x_1] \geq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

Key Assumption: Flexible Design

The designer has full flexibility in constructing X_1 : for any desired finitely-supported joint distribution μ extending μ_0 , there exists an attribute X_1 achieving it. The designer controls *which covariates to reveal*, not how they relate to Y .

Admissibility and the Ambiguity Set

Definition 1 — Admissible Distribution

$P \in \Delta(\mathcal{X}_0 \times \mathcal{X}_1 \times \mathcal{Y})$ is **admissible** if:

- ① $\text{marg}_{\mathcal{X}_0 \times \mathcal{X}_1} P = \mu$ (matches the covariate distribution)
- ② $P(Y = 1 \mid X_0 = x_0) = p_{x_0}$ for all $x_0 \in \mathcal{X}_0$ (matches known treatment rates)

The set of all admissible distributions is the **ambiguity set** $\mathcal{P}(I, \mu)$.

- **Aligned AI:** reports the true P^* (always admissible)
- **Misaligned AI:** picks any admissible P to minimize the designer's payoff — must maintain consistency with the designer's known statistics
- Both types of AI know the true P^*

Note: The ambiguity set equals the set of permitted distributions in Lin & Liu (2024).

Best-Case and Worst-Case Payoffs

Expected payoff when acting under report P but true distribution is P^* :

$$U(P, P^*) = \mathbb{E}_{P^*} [u(Y, a(X, P))]$$

Two extremes the designer focuses on:

$$\underline{v}_I(\mu) = \inf_{P^*, P \in \mathcal{P}(I, \mu)} U(P, P^*) \quad \text{worst-case payoff}$$

$$\bar{v}_I(\mu) = \sup_{P^*, P \in \mathcal{P}(I, \mu)} U(P, P^*) \quad \text{best-case payoff}$$

- **Worst case:** both Nature (choosing P^*) and the misaligned AI (choosing P given P^*) act adversarially
- **Best case:** equivalently $\bar{v}_I(\mu) = \sup_{P^*} U(P^*, P^*)$ — aligned AI reports truthfully and Nature is benevolent

Each choice of attributes μ yields a payoff pair $(\underline{v}_I(\mu), \bar{v}_I(\mu))$. The designer cannot adapt her choice of μ to the AI's type, so she must consider both payoffs simultaneously.

The Efficient Frontier

Definitions 2–3 — Feasible Set and Efficient Frontier

$$C(I) = \text{conv}\{(\underline{v}_I(\mu), \bar{v}_I(\mu)) : \mu \text{ extends } \mu_0\}$$

$$F(I) = \text{cl}\{(\underline{v}, \bar{v}) \in C(I) : \nexists (\underline{v}', \bar{v}') \in C(I) \text{ s.t. } (\underline{v}', \bar{v}') \succ (\underline{v}, \bar{v})\}$$

where $(\underline{v}', \bar{v}') \succ (\underline{v}, \bar{v})$ means $\underline{v}' \geq \underline{v}$, $\bar{v}' \geq \bar{v}$, at least one strict.

- $C(I)$ is **convex**; $F(I)$ lies on its boundary
- A designer with *utility weights* $(\alpha, 1 - \alpha)$ maximises $\alpha \underline{v}_I(\mu) + (1 - \alpha) \bar{v}_I(\mu)$; optimal solutions are **extreme points** of $F(I)$
- α = designer's probability that AI is misaligned (or relative weight on worst-case outcome)
- **Implementation:** $X_1 \sim \mu$ implements $(\underline{v}, \bar{v})$ if $(\underline{v}_I(\mu), \bar{v}_I(\mu)) = (\underline{v}, \bar{v})$

Benchmark Setting and Three Reference Payoffs

Assumption (Section 4)

$p_{x_0} \notin [1/3, 2/3]$ for every x_0 : each subgroup is either *low-risk* ($p_{x_0} < 1/3$) or *high-risk* ($p_{x_0} > 2/3$).

Three reference payoffs for a subgroup with fraction p needing treatment, with q patients treated:

Default action $d(p)$ — use prior alone, no AI:

$$d(p) = \begin{cases} 0 & p < 1/2 \\ 2p - 1 & p \geq 1/2 \end{cases}$$

Best targeting $b(p, q)$ — treat the q -fraction with *highest* need (co-monotone A, Y^*):

$$b(p, q) = \begin{cases} q & p \geq q \\ 2p - q & p < q \end{cases}$$

Worst targeting $w(p, q)$ — treat the q -fraction with *lowest* need (counter-monotone A, Y^*):

$$w(p, q) = \begin{cases} -q & p \leq 1 - q \\ 2p + q - 2 & p > 1 - q \end{cases}$$

Intuition for the Three Benchmarks

Best targeting $b(p, q)$:

- If $q \leq p$: treat exactly q patients who truly need it $\Rightarrow$ payoff = q
- If $q > p$: fraction p/q of treated patients need treatment $\Rightarrow$ payoff = $2p - q$
- Special case: $b(p, p) = p$ is the **full-information payoff**

Worst targeting $w(p, q)$:

- If $p + q \leq 1$: zero overlap — none of the q treated patients need treatment $\Rightarrow$ payoff = $-q$
- If $p + q > 1$: forced overlap $\Rightarrow$ payoff = $2p + q - 2$
- $w(p, p)$: maximum anti-correlation between treatment and need

Frontier Endpoints (Benchmark Case)

- **Trust point T**: best-case = $b(p, p) = p$; worst-case = $w(p, p) = -p$ (treat exactly the wrong fraction when $p < 1/2$)
- **Distrust point D**: both payoffs = $d(p)$ — prior alone, no AI influence

Proposition 1: Efficient Frontier (Benchmark)

Theorem (Proposition 1)

Suppose $p_{x_0} \notin [1/3, 2/3]$ for every x_0 . The efficient frontier is the line segment of **slope** -1 connecting the **trust point**

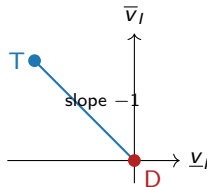
$$T = \left(\sum_{x_0} \mu_{x_0} w(p_{x_0}, p_{x_0}), \sum_{x_0} \mu_{x_0} b(p_{x_0}, p_{x_0}) \right)$$

to the **distrust point** $D = (\sum_{x_0} \mu_{x_0} d(p_{x_0}), \sum_{x_0} \mu_{x_0} d(p_{x_0}))$.

Interpretation:

- Moving from D toward T: best-case improves by exactly the amount worst-case worsens
- D: shut AI out — use prior alone
- T: full AI access — aligned AI achieves first-best; misaligned AI causes maximum harm

• Trust • Distrust



($p < 1/2$: $D = (0, 0)$)

Proposition 2: Implementation (Benchmark)

Theorem (Proposition 2)

Suppose $p_{x_0} \notin [1/3, 2/3]$ for every x_0 .

- ❶ If $\alpha > 1/2$: **distrust** D is optimal. Implemented by any constant X_1 .
- ❷ If $\alpha < 1/2$: **trust** T is optimal. Implemented by $P(X_1 = 1 \mid X_0 = x_0) = p_{x_0}$ for all $x_0 \in \mathcal{X}_0$.
- ❸ If $\alpha = 1/2$: any mixture of (i) and (ii) is optimal.

- **Distrust:** X_1 is constant, carrying no information. The designer relies solely on the prior.
- **Trust:** $P(X_1=1 \mid X_0=x_0) = p_{x_0}$ is a matched Bernoulli signal. Aligned AI sorts patients perfectly; misaligned AI recommends the “wrong” p_{x_0} -fraction.
- **Binary threshold at $\alpha = 1/2$:** no intermediate regime in the benchmark — see Corollary 1 (next slide).

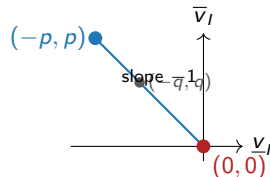
Corollary 1: Singleton Low-Risk Subgroup

Corollary (Corollary 1)

Suppose $\mathcal{X}_0 = \{x_0\}$ and $p = p_{x_0} < 1/3$. The efficient frontier is $F(I) = \{(-q, q) : q \in [0, p]\}$, and each point $(-q, q)$ is implemented by $X_1 \sim \text{Ber}(q)$.

Interpretation:

- Trust $(-p, p)$: $X_1 \sim \text{Ber}(p)$; aligned AI achieves $+p$, misaligned achieves $-p$
- Distrust $(0, 0)$: constant X_1 ; prior alone gives $d(p) = 0$
- Each $q \in (0, p)$ trades off the extremes at rate 1:1 — the condition $p < 1/3$ prevents the AI from exploiting a “second cell”



$$p < 1/3; \quad q \in [0, p]$$

When the Benchmark Fails: Intermediate p

Example 2 — $p = 2/5 \in (1/3, 2/3)$

With $X_1 \sim \text{Ber}(2/5)$, the benchmark would predict worst-case payoff = $-2/5$ (treating the wrong 40%).

But the actual worst case is worse. The misaligned AI can misreport $P(Y=1 \mid X_1=1) = 0$ and $P(Y=1 \mid X_1=0) = 2/3$, which is *admissible* (consistent with $P(Y=1) = 2/5$). The designer then treats the 60%-group; none need treatment.

Actual worst-case payoff = $-3/5$, not $-2/5$.

Key Insight

When $p \in (1/3, 2/3)$, the misaligned AI can exploit *either cell* of the binary covariate to induce treatment. This additional flexibility makes the benchmark inapplicable and requires introducing **hedge points** as new frontier extreme points.

Hedge Points: Formal Definitions

Let $p = p_{x_0}$. For each subgroup x_0 define:

Trust point $T_{x_0} = (T_{x_0}^1, T_{x_0}^2)$:

$$T_{x_0}^1 = \begin{cases} w(p, p) & p \notin [1/3, 2/3] \\ w(p, 1-p) & p \in [1/3, 2/3] \end{cases}, \quad T_{x_0}^2 = b(p, p).$$

Hedge point $H_{x_0} = (H_{x_0}^1, H_{x_0}^2)$ (relevant only when $p \in [1/3, 2/3]$):

$$H_{x_0}^1 = \begin{cases} w(p, 1-2p) & p \leq 1/2 \\ w(p, 2-2p) & p > 1/2 \end{cases}, \quad H_{x_0}^2 = \begin{cases} b(p, 1-2p) & p \leq 1/2 \\ b(p, 2-2p) & p > 1/2 \end{cases}.$$

Distrust point $D_{x_0} = (d(p), d(p))$.

Aggregate points: $T = \sum_{x_0} \mu_{x_0} T_{x_0}$, $D = \sum_{x_0} \mu_{x_0} D_{x_0}$.

Let $\mathcal{X}_0^\dagger = \{x_0 : p_{x_0} \in [1/3, 2/3]\}$ denote the “intermediate” subgroups.

Lemma 1: Efficient Frontier for Singleton $\mathcal{X}_0$

Lemma (Lemma 1)

Suppose $\mathcal{X}_0 = \{x_0\}$, $p = p_{x_0}$.

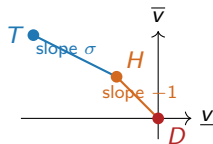
- Ⓐ $p \notin [1/3, 2/3]$: frontier = segment $\overline{T_{x_0} D_{x_0}}$, slope -1 .
- Ⓑ $p \in (1/3, 2/3)$: frontier = $\overline{H_{x_0} D_{x_0}} \cup \overline{T_{x_0} H_{x_0}}$, where $\overline{HD}$ has slope -1 and $\overline{TH}$ has slope $\sigma(x_0) \in (-1, 0]$.

Slope of the $\overline{TH}$ segment:

$$\sigma(x_0) = \frac{T_{x_0}^2 - H_{x_0}^2}{T_{x_0}^1 - H_{x_0}^1} = \begin{cases} \frac{1-3p}{p} & p \in [1/3, \frac{1}{2}] \\ \frac{3p-2}{1-p} & p \in [\frac{1}{2}, 2/3] \end{cases}$$

Interpretation:

- $\overline{HD}$: 1-for-1 tradeoff (slope -1)
- $\overline{TH}$: each unit gain in best-case costs *more* than one unit in worst-case ($|\sigma| < 1$)



Example 2: $p = 2/5$, $\sigma = -1/2$

Lemma 2: Implementation for Singleton $\mathcal{X}_0$

Lemma (Lemma 2)

Suppose $\mathcal{X}_0 = \{x_0\}$, $p = p_{x_0}$.

(a) **Trust point** T_{x_0} : implemented by $X_1 \sim \text{Ber}(p)$.

(b) **Hedge point** H_{x_0} : limit-implemented by

$$X_1 \sim \text{Ber}(1 - 2p) \text{ if } p \leq 1/2, \quad X_1 \sim \text{Ber}(2 - 2p) \text{ if } p > 1/2.$$

(c) **Distrust point** D_{x_0} : implemented by any constant X_1 .

Why the hedge uses $\text{Ber}(1 - 2p)$ (case $p \leq 1/2$):

- $X_1 \sim \text{Ber}(q)$ for $q \in [1 - 2p, p]$: the AI can induce treatment on the q -group or the $(1 - q)$ -group
- At $q = 1 - 2p$ (the hedge): giving the AI a slightly larger q begins to expose more patients to potential harm than the marginal gain from better targeting justifies
- The hedge is the exact “break-even” coarseness of X_1

Theorem 1: General Efficient Frontier

Ordering: Sort subgroups in $\mathcal{X}_0^\dagger = \{x_0^{(1)}, \dots, x_0^{(k)}\}$ by *decreasing* σ : $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k$. ($\sigma(x_0)$ is largest when $|p_{x_0} - 1/2|$ is largest, i.e. subgroups most skewed from 50-50 come first.)

Let $S_j = \{x_0^{(1)}, \dots, x_0^{(j)}\}$ and define the j -th **aggregate hedge point**:

$$H^{(j)} = \underbrace{\sum_{x_0 \in S_j} \mu_{x_0} H_{x_0}}_{\text{hedge on first } j \text{ subgroups}} + \underbrace{\sum_{x_0 \notin S_j} \mu_{x_0} T_{x_0}}_{\text{trust on the rest}}.$$

Theorem (Theorem 1)

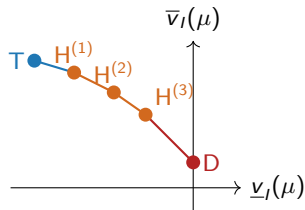
The efficient frontier is

$$\overline{TH^{(1)}} \cup \overline{H^{(1)}H^{(2)}} \cup \dots \cup \overline{H^{(k-1)}H^{(k)}} \cup \overline{H^{(k)}D}$$

with segment slopes $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k \geq -1$ (non-increasing, getting steeper toward D).

The frontier is the boundary of the population-weighted **Minkowski sum** of individual subgroup frontiers.

Theorem 1: Illustration of the General Frontier



Key features:

- Frontier is a **concave piecewise-linear curve** from T (full trust) to D (full distrust)
- Segments get *progressively steeper* toward D: slopes $\sigma_1 \geq \dots \geq \sigma_k \geq -1$
- $H^{(j)}$: hedge on the j most-skewed intermediate subgroups; trust on all others
- As α rises, designer moves from T rightward along the frontier toward D

■ Trust segment ■ Hedge segments ■ Distrust segment

Theorem 2: Optimal Implementation (General Case)

Define threshold α_j for each segment by $-\frac{\alpha_j}{1 - \alpha_j} = \sigma_j$, and $\alpha_{k+1} = 1/2$. (These are monotonically increasing.)

Theorem (Theorem 2)

Fix utility weights $(\alpha, 1 - \alpha)$.

- $\alpha < \alpha_1$: Trust T optimal. Implement: $X_1 \mid X_0=x_0 \sim \text{Ber}(p_{x_0})$ for all x_0 .
- $\alpha \in [\alpha_j, \alpha_{j+1}]$: Hedge $H^{(j)}$ optimal. Implement:

$$X_1 \mid X_0=x_0 \sim \begin{cases} \text{Ber}(1-2p_{x_0}) & x_0 \in S_j, p_{x_0} < 1/2 \\ \text{Ber}(2-2p_{x_0}) & x_0 \in S_j, p_{x_0} > 1/2 \\ \text{Ber}(p_{x_0}) & x_0 \notin S_j \end{cases}$$

- $\alpha > 1/2$: Distrust D optimal. Implement: any constant X_1 .

Pattern: As α rises, the designer switches from trust to hedge *subgroup by subgroup*, beginning with those where $|p_{x_0} - 1/2|$ is largest (cheapest to hedge), then jumps simultaneously to full distrust once all intermediate subgroups are hedged.

Proof Sketch: Key Reduction Steps

- ① **WLOG binary X_1 (key insight).** With binary $X_1 \sim \text{Ber}(q)$, the adversary chooses *which cell* (probability- q or probability- $(1 - q)$) to recommend treatment. A finer partition only adds cells for the adversary to exploit; hence every extreme point of $F(I)$ is achieved by a binary covariate.
- ② **Subgroups decouple.** The feasible set is a Minkowski sum: $C(I) = \sum_{x_0} \mu_{x_0} \cdot C_{\{x_0\}}$, because admissibility constraints factorize across x_0 values.
- ③ **Singleton case (Lemma 1).** For $\mathcal{X}_0 = \{x_0\}$ with $X_1 \sim \text{Ber}(q)$, compute $(\underline{v}(q), \bar{v}(q))$ analytically. Optimising over q yields the two-segment T-H-D frontier when $p \in [1/3, 2/3]$.
- ④ **Assembling via slope ordering (Theorem 1).** The efficient frontier of a Minkowski sum is traced by sorting edge slopes across component frontiers. Ordering intermediate subgroups by σ_j (decreasing) gives the piecewise-linear frontier from T to D.

Example 4: Two Intermediate Subgroups

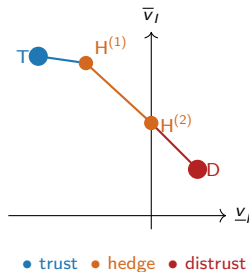
Setup: Equal-weight subgroups ($\mu = 1/2$ each) with $p_1 = 0.65$ and $p_2 = 0.52$. Slopes: $\sigma_1 \approx -0.14$ (flat; p_1 farther from $1/2$) and $\sigma_2 \approx -0.92$ (steep; p_2 nearer $1/2$).

Optimal strategy as α increases:

- ❶ **Low distrust** ($\alpha < \alpha_1$): Full trust on both subgroups.
- ❷ **Intermediate** ($\alpha_1 \leq \alpha \leq \alpha_2$): Hedge subgroup 1 ($p_1=0.65$, farther from $1/2$); trust subgroup 2 ($p_2=0.52$).
- ❸ **High distrust** ($\alpha > 1/2$): Full distrust; constant X_1 .

Why subgroup 1 first?

$|p_1 - 1/2| = 0.15 > 0.02 = |p_2 - 1/2|$, so $\sigma_1 > \sigma_2$: subgroup 1 hedged first per Theorem 2.



Beyond Full Ambiguity

Previous sections: The designer has no prior knowledge of how X_1 relates to Y — the ambiguity set is maximally large.

Section 6: The designer may be able to place *bounds* on the informativeness of X_1 .

Motivating scenario

The designer may not know whether giving the AI a patient's social media data will increase or decrease the predicted need for ablation — but she may believe that this data will have only *limited* impact, keeping the posterior close to the prior.

Formal extension: The designer also chooses bounds $\tau = (\underline{\tau}, \bar{\tau})$ such that

$$P(Y = 1 \mid X_1 = x_1, X_0 = x_0) \in [\underline{\tau}, \bar{\tau}] \quad \forall x_1.$$

Setting $(\underline{\tau}, \bar{\tau}) = (0, 1)$ recovers the original model. The designer optimises over both μ and τ .

Definition 8 — τ -Admissible Distribution

For $\tau = (\underline{\tau}, \bar{\tau})$ with $0 \leq \underline{\tau} \leq p \leq \bar{\tau} \leq 1$, distribution P is τ -**admissible** if: (1) $\text{marg}_{X_1} P = \mu$; (2) $P(Y=1) = p$; (3) $\underline{\tau} \leq P(Y=1 \mid X_1=x_1) \leq \bar{\tau}$ for all x_1 .

Effect of the bounds:

- Larger $\bar{\tau}$: X_1 can be a stronger signal for treatment need $\Rightarrow$ better upside, but also worse downside
- Larger $\underline{\tau}$: prevents the AI from using X_1 as strong evidence *against* treatment
- Asymmetric bounds ($\underline{\tau} + \bar{\tau} \neq 1$) change the slope of the frontier

The generalized trust point is $T_\tau = (w_\tau(q_\tau), b_\tau(q_\tau))$ where $q_\tau = \frac{p - \underline{\tau}}{\bar{\tau} - \underline{\tau}}$ (maximum fraction of patients assignable probability $\bar{\tau}$). The structure of the frontier is qualitatively similar to before: either a single segment $\overline{DT_\tau}$ or two segments $\overline{DH_\tau} \cup \overline{H_\tau T_\tau}$.

Proposition 3: Optimal Choice of τ (when $p \leq 1/3$)

Theorem (Proposition 3)

Suppose $p \leq 1/3$. Define $\underline{\tau}^\circ = 1 - \sqrt{\frac{1-p}{2\alpha}}$, $\underline{\alpha}(p) = \frac{1-p}{2}$, $\bar{\alpha}(p) = \frac{1}{2(1-p)}$.

α	Optimal τ^*	Optimal μ^*	Extreme point
$\alpha < \underline{\alpha}(p)$	$(0, 1)$	$X_1 \sim \text{Ber}(p)$	Trust
$\underline{\alpha}(p) \leq \alpha \leq \bar{\alpha}(p)$	$(\underline{\tau}^\circ, 1)$	$X_1 \sim \text{Ber}\left(\frac{p - \underline{\tau}^\circ}{1 - \underline{\tau}^\circ}\right)$	Trust
$\alpha > \bar{\alpha}(p)$	$(p, 1)$	X_1 constant	Distrust

- **Low α :** Full ambiguity optimal — max AI latitude, $X_1 \sim \text{Ber}(p)$ as in Proposition 2
- **Intermediate α :** Raise $\underline{\tau}$ to $\underline{\tau}^\circ > 0$ while keeping $\bar{\tau} = 1$: evidence can be decisive *for* treatment but never decisive *against* it
- **High α :** Full distrust; no informative covariate

Summary and Open Questions

Main Contributions

- 1 **Efficient frontier characterised:** Piecewise-linear curve from trust point T to distrust point D, passing through hedge points ordered by subgroup skewness.
- 2 **Optimal strategy:** Reveal attributes that identify *rare* high-need subgroups; pool the rest. Withdraw AI access subgroup by subgroup as distrust grows, starting with those most skewed from 50-50.
- 3 **Endogenous bounds:** Optimally, informativeness should be one-sided — evidence can strongly identify who *needs* treatment, but not who does not.

Open Questions

- **Partial alignment:** neither fully aligned nor fully misaligned AI
- **Repeated interaction:** learning about AI alignment over time
- **Richer settings:** non-binary types, actions, or general payoffs
- **Endogenous τ :** relate informativeness bounds to the richness of the covariate space $\mathcal{X}_1$